

On the Resilience Order of Weightwise Almost Perfectly Balanced Functions

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Abstract. The recent development of Fully Homomorphic Encryption (FHE) witnessed the emergence of a new generation of tailored cryptographic primitives designed to meet its specific criteria. Among promising candidates for FHE constructions stands out the FLIP cipher, which employs Boolean functions that are evaluated only on specific subsets of $\mathbb{F}_2^n$. In this article, we study Weightwise Almost Perfectly Balanced (WAPB) functions, which are almost balanced on each of these subsets. While WAPB functions have been of great interest for new constructions recently, some aspects, such as resilience remain poorly understood. As such, we take a first step at characterizing the resilience of WAPB functions, through their properties as correctors. We highlight its close connection with the restricted Walsh transform and uncover an algebraic relation between Krawtchouk matrices and Vandermonde matrices, which reduces the problem of determining the corrector order of a WAPB function to a particular instance of the Prouhet-Tarry-Escott problem. This reduction helps us show that for infinitely many integers n , WAPB functions in n variables have corrector order tightly upper bounded by the Hamming weight of n minus one. We conjecture that this observation holds for any positive integer n , which is verified for n up to 62.

Keywords: WAPB function · resilience order · Krawtchouk polynomial · Vandermonde matrix · Prouhet-Tarry-Escott problem

1 Introduction

With the design of FLIP [33], established cryptographic criteria for Boolean functions, which are traditionally used to assess the security of filtered Linear Feedback Shift Registers (LFSRs), combined LFSRs, or more general stream ciphers, no longer directly apply. In this cipher, the relevant properties of the filtering Boolean function are not defined on the entire $\mathbb{F}_2^n$ but instead on a specific, publicly known subset of inputs determined by the execution of the cipher. In particular, the relevant properties of the function are only required on these subsets. This shift has triggered a generalization of standard attacks and the development of new criteria for Boolean functions restricted to subsets or partitions of $\mathbb{F}_2^n$, beginning with the study of restricted cryptographic criteria [7]. So far, these criteria have been regarded for general subsets [4, 7], subsets of fixed Hamming weight e.g. [7, 14, 29, 37], and affine spaces [6].

In FLIP, the inputs to the filtering function always have a fixed and known Hamming weight, and tailored attacks to this constraint strengthens them considerably. This phenomenon reflects a broader trend: additional information about a cipher’s internal values may dramatically simplify attacks, as observed in algebraic side-channel cryptanalysis [39] or lattice reduction with side information [8]. From the FLIP analysis emerged a systematic study of the partition of $\mathbb{F}_2^n$ into $n + 1$ slices, extending the analysis from a single relevant slice to all slices simultaneously, where a slice is the set of all vectors of a given Hamming weight, i.e., $E_{k,n} = \{x \in \mathbb{F}_2^n \mid w_H(x) = k\}$

for $k = 0, 1, \dots, n$. Notably, Hamming weight is also a widely used leakage model in side-channel attacks [13, 23, 41], even though in practice approximate weight leakages are often considered more realistic [2, 20, 38].

A fundamental security requirement for Boolean functions used in stream ciphers is balancedness: the output should take each value equally often to avoid exploitable statistical biases. In the context of slices, [7] introduced Weightwise Perfectly Balanced (WPB) functions and Weightwise Almost Perfectly Balanced (WAPB) functions, which are balanced and almost balanced on every slice $E_{k,n}$, respectively. The qualifier “almost” accounts for the case when n is not a power of two where the slices always have odd cardinality (which prevents perfect balancedness on any slice). In the FLIP cipher, the filtering function is evaluated on inputs of a fixed and known Hamming weight, corresponding to a single slice of $\mathbb{F}_2^n$ (typically of weight close to $n/2$). Accordingly, the cryptographic behaviour of the function on this slice is of primary importance. It is worth noting that the WAPB property is a stronger requirement, as it imposes (almost) balancedness on all slices simultaneously. Such a strengthening can be seen as an extremal or canonical condition (in the similar spirit as bent functions are studied as maximally nonlinear objects for potential cryptographic applications). Moreover, considering several slices can be relevant beyond the nominal setting of cryptanalysis. Certain attacks may involve restrictions of the function or inputs whose Hamming weights deviate from the central slice. For instance, in the case of FLIP, if some variables are not involved in the filtering function, guess-and-determine strategies lead to the analysis of subfunctions whose inputs lie in neighboring slices. More generally, leakage models such as Hamming weight measurements naturally provide information across multiple slices. The aforementioned aspects motivate the study of functions whose properties remain controlled on a range of slices, rather than on a single one.

Since 2017, many works have introduced new constructions of WPB or WAPB functions, studying their global and restricted cryptographic parameters. Recursive families and secondary constructions were first presented in [7]. Constructions based on field representations yield 2-rotation symmetric WPB functions with high weightwise nonlinearity [26, 36], and recently 2- π symmetric WAPB functions for all n [12]. WPB functions obtained by comparing the weights of two halves of the input achieve optimal algebraic immunity [35, 42]. Comparing the two halves using total orders leads to WAPB functions for all n [32]. Other approaches modify near-WPB linear or quadratic functions to impose balancedness on all slices [34]. Such a general idea was later generalized to obtain WAPB functions from low-degree functions [19, 25, 46, 47, 48]. Secondary constructions have also been explored, such as direct sums [49], Siegenthaler constructions combined with symmetric functions [15] and t -concatenation [22]. Moreover, several works have modified the support of non-WPB functions to provide WPB or WAPB functions with bounded parameters for the weightwise nonlinearity [14], the global nonlinearity [17], or the algebraic immunity [16]. Heuristic search approaches, notably evolutionary algorithms, have been applied as well [30, 31, 45]. Additional constructions appear in [10, 11, 22].

Despite this growing body of work, one central cryptographic parameter has remained largely unexplored for WAPB functions: resilience. Generalizing the notion of balancedness, a Boolean function is said t -resilient if it is uncorrelated to all affine functions combining at most t variables. The resilience has been introduced to estimate the complexity of correlation attacks on stream ciphers [40], and remains relevant when analyzing Boolean functions under restricted input distributions. Previous studies on WAPB functions have characterized their nonlinearity [17], restricted nonlinearity [14], algebraic immunity (both global [16] and weightwise [4]), algebraic degree [18], yet no general result on the resilience of WAPB functions had been established.

The resilience of a WAPB function is particularly intriguing since it is related to both global balancedness and balancedness on all the parts of a particular partition. Since the slices and the hyperplanes corresponding to affine relations are not related, one might expect that enforcing balancedness on all slices would not significantly constrain other properties such as resilience. However, our results show that this intuition does not hold. Even the simplest case of WPB functions provides a stark example: although they are balanced on every slice, they cannot be 1-resilient. More generally, WAPB functions have a low resilience order (the maximum value of t such that the function is t -resilient). We can exhibit n -variable WAPB functions with a resilience order equal to $w_H(n) - 1$, and demonstrate for infinite values of n that no WAPB function can be $w_H(n)$ -resilient.

To analyze the resilience of WAPB functions, we adopt a perspective that is uncommon in Boolean functions used for cryptography, we study their properties as correctors e.g. [21, 24, 27] since the corrector order provides an upper bound on the resilience order. A t -corrector is a function for which the sum of Walsh coefficients over each slice $E_{k,n}$ for $k = 0, 1, \dots, t$ is null. We show (Theorem 1) that the corrector order of a Boolean function is tightly connected to a system of linear equations involving the restricted Walsh transforms on the slices of the function. A key component of our study is an algebraic connection between Krawtchouk matrices and Vandermonde matrices (Theorem 2), which enables us to transform the system of linear equations into a simpler one (Theorem 3) that is associated with a particular Vandermonde matrix. The link between correctors and restricted Walsh transform provides a general framework to study resilience through corrector properties, and is of independent interest for investigating the number and structure of t -corrector Boolean functions. Furthermore, for WAPB functions, those equations are highly simplified since the restricted Walsh transform of these functions takes only the value $-1, 0$ or 1 . This feature allows us to reduce the determination of the maximum corrector order (and hence an upper bound on the resilience) of WAPB functions to a particular instance of the Prouhet-Tarry-Escott problem (see e.g. [5]). Our last main result (Theorem 4) shows that for infinitely many values of n , n -variable WAPB functions have corrector order upper bounded by $w_H(n) - 1$. The existence of WAPB functions with resilience $w_H(n) - 1$ confirms the tightness of this upper bound. Following experimental verification for n up to 62 (as well as for other integers n with $w_H(n)$ up to 5), we propose a conjecture (Conjecture 1) that all n -variable WAPB functions have corrector order tightly upper bounded by $w_H(n) - 1$.

2 Preliminaries

Let $\mathbb{N}$ be the set of all non-negative integers and $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$. For a positive integer n , we denote by $[n]$ the set of all integers from 1 to n , i.e. $\{1, \dots, n\}$ and use the notation $[0, n]$ for the set $\{0, 1, \dots, n\}$. For readability, we use the notation $+$ instead of $\oplus$ for addition in $\mathbb{F}_2$. For a vector $v \in \mathbb{F}_2^n$, we denote its Hamming weight by $w_H(v)$, defined as $w_H(v) = |\{i \in [n] \mid v_i = 1\}|$. The inner product between two vectors $u, v \in \mathbb{F}_2^n$ is given by $u \cdot v = \sum_{i=1}^n u_i v_i$. For $k \in [0, n]$, the k -th slice of $\mathbb{F}_2^n$ is the set $E_{k,n} = \{x \in \mathbb{F}_2^n \mid w_H(x) = k\}$. The support of a positive integer a in binary expression is defined as $\text{supp}(a) = \{i \mid \sum_{i \geq 0} 2^i = a\}$. For two integers $a, b \in \mathbb{N}$, we denote by $a \preceq b$ when $\text{supp}(a) \subseteq \text{supp}(b)$.

2.1 Boolean Functions and Cryptographic Properties

The set of all Boolean functions from $\mathbb{F}_2^n$ to $\mathbb{F}_2$ will be denoted by $\mathcal{B}_n$.

Definition 1 (Balancedness). A Boolean function $f \in \mathcal{B}_n$ is said to be balanced if $|\text{supp}(f)| = 2^{n-1} = |\text{supp}(f+1)|$, where $\text{supp}(f)$ denotes the support of f , namely, the set $\{x \in \mathbb{F}_2^n \mid f(x) = 1\}$.

Definition 2 (Walsh transform). Let $f \in \mathcal{B}_n$ be a Boolean function. Its Walsh transform W_f at $a \in \mathbb{F}_2^n$ is defined as:

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)+a \cdot x}.$$

Definition 3 (Resilience). A Boolean function $f \in \mathcal{B}_n$ is said to be t -resilient if for all $a \in \mathbb{F}_2^n$ such that $0 \leq \text{wt}_H(a) \leq t$ we have $W_f(a) = 0$. We denote by $\text{res}(f)$ the maximum value of t such that f is t -resilient.

Definition 4 (Corrector). A Boolean function $f \in \mathcal{B}_n$ is said to be a t -corrector if for all $0 \leq k \leq t$ it satisfies:

$$\sum_{a \in \mathbf{E}_{k,n}} W_f(a) = 0.$$

We denote by $\text{corr}(f)$ the maximum value of t such that f is a t -corrector.

According to the definitions, it is clear that $\text{corr}(f) \geq \text{res}(f)$ for any $f \in \mathcal{B}_n$. We shall use this fact to prove the non-existence of t -resilient WAPB functions by investigating the maximum corrector order of WAPB functions, which will be introduced in the next subsection.

2.2 Restricted Walsh Transform and WAPB Functions

The Boolean hypercube is partitioned into $n+1$ slices, where each slice consists of vectors sharing the same Hamming weight. We refer to properties that hold slice-wise as weightwise properties. The n -variable symmetric Boolean functions are precisely those that are constant on each slice.

Definition 5 (Restricted Walsh transform). Let $f \in \mathcal{B}_n$, $S \subset \mathbb{F}_2^n$, its Walsh transform restricted to S at $a \in \mathbb{F}_2^n$ is defined as:

$$W_{f,S}(a) = \sum_{x \in S} (-1)^{f(x)+a \cdot x}.$$

For $S = \mathbf{E}_{\ell,n}$ we denote $W_{f,\mathbf{E}_{\ell,n}}(a)$ by $\mathcal{W}_{f,\ell}(a)$, and for $a = 0_n$ we denote $\mathcal{W}_{f,\ell}(a)$ by $\mathcal{W}_{f,\ell}$.

Definition 6. For $f \in \mathcal{B}_n$, we define its restricted Walsh vector in $\mathbb{Z}^{n+1}$ as

$$\text{rw}_f = (\mathcal{W}_{f,0}, \mathcal{W}_{f,1}, \dots, \mathcal{W}_{f,n-1}, \mathcal{W}_{f,n})^\top.$$

Definition 7 (Weightwise (Almost) Perfectly Balanced Function [7]). Let $m \in \mathbb{N}^*$ and f be a Boolean function in $n = 2^m$ variables. It will be called weightwise perfectly balanced (WPB) if f is balanced on $\mathbf{E}_{\ell,n}$ for every $\ell \in [1, n-1]$ and

$$f(0, \dots, 0) = 0, \quad \text{and} \quad f(1, \dots, 1) = 1.$$

The set of WPB functions in 2^m variables is denoted by WPB_m .

When n is not a power of 2, some other weights than $\ell = 0$ and n give slices of odd cardinality. We consider the generalization for any $n \in \mathbb{N}^*$, in this case we call $f \in \mathcal{B}_n$ weightwise almost perfectly balanced (WAPB) if:

$$|\text{supp}_\ell(f)| = \begin{cases} |\mathbf{E}_{\ell,n}|/2 & \text{if } |\mathbf{E}_{\ell,n}| \text{ is even,} \\ (|\mathbf{E}_{\ell,n}| \pm 1)/2 & \text{if } |\mathbf{E}_{\ell,n}| \text{ is odd.} \end{cases}$$

The set of WAPB functions in n variables is denoted by $\mathcal{WAPB}_n$.

Below we recall Lucas's Theorem.

Lemma 1. For a prime p and non-negative integers n, k with $n \geq k$, the following congruence relation holds:

$$\binom{n}{k} \equiv \prod_{i=0}^r \binom{n_i}{k_i} \pmod{p},$$

where $n = n_r p^r + \dots + n_1 p + n_0$ and $k = k_r p^r + \dots + k_1 p + k_0$ are the p -ary expansions of n and k , respectively.

This implies that $|\mathbf{E}_{\ell,n}| = \binom{n}{\ell}$ is odd if and only if $\ell \preceq n$, i.e., $\text{supp}(\ell) \subseteq \text{supp}(n)$. By the definition of WAPB functions, we easily have the following property of WAPB functions:

Property 1 (WAPB functions and restricted Walsh transform). Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_n$ is WAPB if and only if:

$$\forall \ell \in [0, n], \mathcal{W}_{f,\ell} = \begin{cases} 0 & \text{if } \ell \not\preceq n, \\ \pm 1 & \text{if } \ell \preceq n. \end{cases}$$

In particular, when $n = 2^m$ with $m \in \mathbb{N}^*$, $f \in \mathcal{B}_n$ is WPB if and only if:

$$\mathcal{W}_{f,0} = 1, \quad \mathcal{W}_{f,n} = -1, \quad \text{and} \quad \forall \ell \in [1, n-1], \mathcal{W}_{f,\ell} = 0.$$

We make use of Krawtchouk polynomials to characterize the corrector order of WAPB functions. We give the necessary preliminaries here and refer to e.g. [28, Ch.5, §7] for more details.

Definition 8 (Krawtchouk Polynomials). The Krawtchouk polynomial of degree k , with $0 \leq k \leq n$ is given by:

$$\mathsf{K}_k(\ell, n) = \sum_{j=0}^k (-1)^j \binom{\ell}{j} \binom{n-\ell}{k-j},$$

where the binomial coefficient $\binom{m}{r} = 0$ when $m < r$ or $r < 0$ by convention.

Krawtchouk polynomials can be characterized by the generating series:

$$(1+z)^{n-\ell}(1-z)^\ell = \sum_{k=0}^{\infty} \mathsf{K}_k(\ell, n) z^k.$$

In addition, there is an interesting connection between Krawtchouk polynomials and linear functions restricted on the slice $\mathbf{E}_{k,n}$:

$$\mathsf{K}_k(\ell, n) = \sum_{u \in \mathbf{E}_{k,n}} (-1)^{u \cdot v}, \quad \forall v \in \mathbf{E}_{\ell,n}. \quad (1)$$

Property 2 (Krawtchouk polynomials relation). Let $n \in \mathbb{N}^*$ and $k \in [0, n]$. The following relations hold:

- $K_k(n - \ell, n) = (-1)^k K_k(\ell, n)$,
- $K_{n-k}(x, n) = (-1)^x K_k(x, n)$,
- if n is even and k is odd, $K_k(n/2, n) = 0$,
- if n is even, $K_{n/2}(1, n) = 0$.
- $\binom{n}{\ell} K_k(\ell, n) = \binom{n}{k} K_\ell(k, n)$.

Property 3 ([9, Prop.4]). Let $n \in \mathbb{N}^*$ and $k \in [0, n]$, the following hold:

$$(k+1)K_{k+1}(\ell, n) = (n-2\ell)K_k(\ell, n) - (n-k+1)K_{k-1}(\ell, n).$$

3 Characterizations of t -correctors

The following theorem characterizes the t -corrector property in terms of the restricted Walsh transform. This allows us to reformulate WAPB functions of corrector order t in terms of Krawtchouk polynomials. Furthermore, by Theorem 2, the following characterization will be converted to a linear system defined by a Vandermonde matrix, which facilitates our investigation of corrector orders of WAPB functions.

Theorem 1. *Let $n \in \mathbb{N}^*$ and $t \in \mathbb{N}$ such that $t \leq n$. A Boolean function $f \in \mathcal{B}_n$ has $\text{corr}(f) = t$ if and only if t is the maximum integer such that the restricted Walsh transform satisfies*

$$\sum_{\ell=0}^n \mathcal{W}_{f,\ell} K_k(\ell, n) = 0 \quad \text{for } k = 0, 1, \dots, t. \quad (2)$$

Proof. Note that

$$\begin{aligned} \sum_{a \in E_{k,n}} W_f(a) &= \sum_{a \in E_{k,n}} \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + a \cdot x} = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)} \left(\sum_{a \in E_{k,n}} (-1)^{a \cdot x} \right) \\ &= \sum_{\ell=0}^n \sum_{x \in E_{\ell,n}} (-1)^{f(x)} \left(\sum_{a \in E_{k,n}} (-1)^{a \cdot x} \right) = \sum_{\ell=0}^n \sum_{x \in E_{\ell,n}} (-1)^{f(x)} K_k(\ell, n) \\ &= \sum_{\ell=0}^n \mathcal{W}_{f,\ell} K_k(\ell, n), \end{aligned}$$

where the last second equality follows from (1). The desired statement is then proved by Definition 4. $\square$

Let $n \in \mathbb{N}^*$ and $t \in \mathbb{N}$ such that $t \leq n$, we define the Krawtchouk matrix $\mathbf{K}$ as an $(n+1) \times (n+1)$ matrix with entries given by

$$\forall k \in [0, n], \ell \in [0, n], \quad \mathbf{K}_{k,\ell} = K_k(\ell, n).$$

Denote by $\mathbf{K}^{(t+1)}$ the submatrix formed by the leading $(t+1)$ rows of $\mathbf{K}$. By Theorem 1 we see that f is a t -corrector if and only if:

$$\mathbf{K}^{(t+1)} \cdot \text{rw}_f = \mathbf{0}_{t+1}, \quad (3)$$

where $\mathbf{0}_{t+1}$ denotes the column vector of $(t+1)$ zeros. The following theorem provides an interesting algebraic relation between the above Krawtchouk matrix $\mathbf{K}$ and the Vandermonde matrix defined by $0, 1, \dots, n$ via a non-singular lower triangular matrix³. This connection facilitates the study of the above equation in (3).

Theorem 2. *Let*

$$\alpha_k = -\frac{n-k}{2}, \beta_k = \frac{n}{2}, \gamma_k = -\frac{k}{2}, \quad 0 \leq k \leq n,$$

and for $1 \leq r \leq n$ define $r \times (r+1)$ matrices $\mathbf{H}_r$ as follows:

$$\mathbf{H}_r = \begin{bmatrix} \beta_0 & \gamma_1 & & & \\ \alpha_0 & \beta_1 & \gamma_2 & & \\ & \alpha_1 & \beta_2 & \gamma_3 & \\ & & \ddots & \ddots & \ddots \\ & & & \alpha_{r-2} & \beta_{r-1} & \gamma_r \end{bmatrix}. \quad (4)$$

Let $\mathbf{K}$ be the Krawtchouk matrix with $\mathbf{K}_{k,\ell} = K_k(\ell, n)$ and $\mathbf{V}$ be the Vandermonde matrix with $V_{k,\ell} = \ell^k$, where k, ℓ with $0 \leq k, \ell \leq n$ index the rows and columns, respectively. Then the matrix $\mathbf{M}$ satisfying

$$\mathbf{M} \cdot \mathbf{K} = \mathbf{V},$$

is an invertible lower triangular matrix whose rows are given by the recursion:

$$\mathbf{M}_{0,0} = 1, \quad \mathbf{M}_{r,[0,r]} = \mathbf{M}_{r-1,[0,r-1]} \mathbf{H}_r \quad \text{for } r = 1, 2, \dots, n, \quad (5)$$

where $\mathbf{M}_{r,[0,r]} = [\mathbf{M}_{r,0}, \dots, \mathbf{M}_{r,r}]$ denotes the leading $(r+1)$ entries in the r -th row of $\mathbf{M}$; in other words, $\mathbf{M}_{r,[0,r]} = [1] \cdot \mathbf{H}_1 \cdot \mathbf{H}_2 \cdots \mathbf{H}_r$, $r = 2, \dots, n$.

Proof. Note that both the Krawtchouk matrix and the Vandermonde matrix invertible by definition. Property 3 of the Krawtchouk polynomial is crucial for us to show that the matrix $\mathbf{M}$ satisfying $\mathbf{M} \cdot \mathbf{K} = \mathbf{V}$ is an invertible lower triangular matrix.

We will proceed by induction on r to gradually determine the non-zero entries in each row of $\mathbf{M}$ denoted as the vector $\mathbf{M}_{r,[0,r+1]}$. Property 3 can be equivalently stated as

$$\ell \cdot K_k(\ell, n) = -\frac{n-(k-1)}{2} K_{k-1}(\ell, n) + \frac{n}{2} K_k(\ell, n) - \frac{k+1}{2} K_{k+1}(\ell, n),$$

which can be rewritten as

$$\begin{aligned} \ell \cdot K_k(\ell, n) &= \alpha_{k-1} K_{k-1}(\ell, n) + \beta_k K_k(\ell, n) + \gamma_{k+1} K_{k+1}(\ell, n) \\ &= \begin{bmatrix} \alpha_{k-1} & \beta_k & \gamma_{k+1} \end{bmatrix} \cdot \begin{bmatrix} K_{k-1}(\ell, n) \\ K_k(\ell, n) \\ K_{k+1}(\ell, n) \end{bmatrix}, \end{aligned} \quad (6)$$

³ Surprisingly, it appears that the relation between these two well-studied matrices has not been documented in the literature.

where $\alpha_{k-1} = -\frac{n-(k-1)}{2}$, $\beta_k = \frac{n}{2}$ and $\gamma_{k+1} = -\frac{k+1}{2}$ as defined in Theorem 2.

We start with the case of $r = 0$. Since $K_0(\ell, n) = 1$ for all ℓ, n , it follows that $\mathbf{M}_{0,[0,n]} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}$ is lower triangular and verifies $\mathbf{M}_{0,[0,n]} \cdot \mathbf{K} = \mathbf{V}_{0,[0,n]}$.

Now we move to $r = 1$. In this case we need to determine $\mathbf{M}_{1,[0,2]}$ that yields ℓ^1 , which corresponds to Row 1 in $\mathbf{V}$. We can expand ℓ^1 using Eq. (6) as follows:

$$\begin{aligned} \ell^1 &= \ell \cdot K_0(\ell, n) \\ &= \beta_0 K_0(\ell, n) + \gamma_1 K_1(\ell, n) \\ &= \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{\mathbf{M}_{0,[0,1]}} \cdot \underbrace{\begin{bmatrix} \beta_0 & \gamma_1 \end{bmatrix}}_{\mathbf{H}_1} \cdot \begin{bmatrix} K_0(\ell, n) \\ K_1(\ell, n) \end{bmatrix}. \end{aligned}$$

As such, $\mathbf{M}_{1,[0,2]} = \begin{bmatrix} \beta_0 & \gamma_1 \end{bmatrix}$ and $\mathbf{M}_{1,[0,n]} \cdot \mathbf{K} = \mathbf{V}_{1,[0,n]}$. The second row also confirms that $\mathbf{M}$ is lower triangular.

Let $r = 2$. We want to determine $\mathbf{M}_{2,[0,3]}$ that yields ℓ^2 in $\mathbf{V}$. Similarly, we expand ℓ^2 using Eq. (6) as follows:

$$\begin{aligned} \ell^2 &= \ell \cdot \ell = \ell \left(\gamma_1 K_1(\ell, n) + \beta_0 K_0(\ell, n) \right) \\ &= \beta_0 \ell K_0(\ell, n) + \gamma_1 \ell K_1(\ell, n) \\ &= \beta_0 \begin{bmatrix} \beta_0 & \gamma_1 \end{bmatrix} \cdot \begin{bmatrix} K_0(\ell, n) \\ K_1(\ell, n) \end{bmatrix} + \gamma_1 \begin{bmatrix} \alpha_0 & \beta_1 & \gamma_2 \end{bmatrix} \cdot \begin{bmatrix} K_0(\ell, n) \\ K_1(\ell, n) \\ K_2(\ell, n) \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} \beta_0 & \gamma_1 \end{bmatrix}}_{\mathbf{M}_{1,[0,2]}} \cdot \underbrace{\begin{bmatrix} \beta_0 & \gamma_1 & 0 \\ \alpha_0 & \beta_1 & \gamma_2 \end{bmatrix}}_{\mathbf{H}_2} \cdot \begin{bmatrix} K_0(\ell, n) \\ K_1(\ell, n) \\ K_2(\ell, n) \end{bmatrix}. \end{aligned}$$

As such, $\mathbf{M}_{2,[0,3]} = \mathbf{M}_{1,[0,2]} \cdot \mathbf{H}_2 = \left[(\beta_0^2 + \gamma_1 \alpha_0) \ (\gamma_1 \beta_0 + \gamma_1 \beta_1) \ (\gamma_1 \gamma_2) \right]$ is lower triangular. Thus, we have

$$\mathbf{M}_{2,[0,n]} \cdot \mathbf{K} = \mathbf{M}_{2,[0,3]} \begin{bmatrix} K_0(\ell, n) \\ K_1(\ell, n) \\ K_2(\ell, n) \end{bmatrix} = \mathbf{V}_{2,[0,n]}.$$

Induction: we shall prove that if the statement is true for $r - 1$, then it is true for r .

Again, we use Eq. (6) to expand ℓ^r as follows:

$$\begin{aligned}
\ell^r &= \ell \cdot \ell^{r-1} = \ell \left(\sum_{k=0}^{r-1} \mathbf{M}_{r-1,i} \cdot \mathbf{K}_k(\ell, n) \right) \\
&= \begin{bmatrix} \mathbf{M}_{r-1,0} & \mathbf{M}_{r-1,1} & \dots & \mathbf{M}_{r-1,r-1} \end{bmatrix} \cdot \begin{bmatrix} \ell \mathbf{K}_0(\ell, n) \\ \ell \mathbf{K}_1(\ell, n) \\ \vdots \\ \ell \mathbf{K}_{r-1}(\ell, n) \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{M}_{r-1,0} & \mathbf{M}_{r-1,1} & \dots & \mathbf{M}_{r-1,r-1} \end{bmatrix} \cdot \begin{bmatrix} \beta_0 & \gamma_1 & & \\ \alpha_0 & \beta_1 & \gamma_2 & \\ & \ddots & \ddots & \ddots \\ & & \alpha_{r-2} & \beta_{r-1} & \gamma_r \end{bmatrix} \cdot \begin{bmatrix} \mathbf{K}_0(\ell, n) \\ \mathbf{K}_1(\ell, n) \\ \vdots \\ \mathbf{K}_r(\ell, n) \end{bmatrix} \\
&= \mathbf{M}_{r-1,[0,r]} \cdot \mathbf{H}_r \cdot \begin{bmatrix} \mathbf{K}_0(\ell, n) \\ \mathbf{K}_1(\ell, n) \\ \vdots \\ \mathbf{K}_r(\ell, n) \end{bmatrix},
\end{aligned}$$

which implies $\mathbf{M}_{r,[0,r]} = \mathbf{M}_{r-1,[0,r]} \cdot \mathbf{H}_r$ and $\mathbf{M}_{r,[0,n]} \cdot \mathbf{K} = \mathbf{V}_{r,[0,n]}$, and that $\mathbf{M}$ is lower triangular. The desired statement follows. $\square$

Denote by $\mathbf{M}_{[0,t],[0,t]}$ the $(t+1)$ -order submatrix formed by the leading $(t+1)$ rows and columns of $\mathbf{M}$. The matrix $\mathbf{M}_{[0,t],[0,t]}$ is invertible since the matrix $\mathbf{M}$ is an invertible lower triangular matrix by Theorem 2. The fact that $\mathbf{M} \cdot \mathbf{K} = \mathbf{V}$ with $\mathbf{M}$ being an invertible lower triangular matrix implies that

$$\mathbf{M}_{[0,t],[0,t]} \cdot \mathbf{K}^{(t+1)} = \mathbf{V}^{(t+1)},$$

where $\mathbf{V}^{(t+1)}$ is the submatrix formed by the leading $(t+1)$ rows of $\mathbf{V}$. The above equality together with (3) enables us to reformulate Theorem 1 as follows.

Theorem 3. *Let $n \in \mathbb{N}^*$ and $t \in \mathbb{N}$ such that $t \leq n$. A Boolean function $f \in \mathcal{B}_n$ has $\text{corr}(f) = t$ if and only if t is the maximum integer such that the restricted Walsh transform vector rw_f satisfies*

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 2 & \dots & n \\ & & \dots & \dots & \\ 0 & 1 & 2^t & \dots & n^t \end{bmatrix} \cdot \begin{bmatrix} \mathcal{W}_{f,0} \\ \mathcal{W}_{f,1} \\ \vdots \\ \mathcal{W}_{f,n} \end{bmatrix} = \mathbf{0}_{t+1}. \quad (7)$$

Furthermore, due to Property 1 that any WAPB function f has its restricted Walsh transform $\mathcal{W}_{f,\ell} = 0$ when $\ell \not\leq n$, it suffices to investigate the values of $\mathcal{W}_{f,\ell}$ when $\ell \leq n$. Assume $\mathbf{w}_H(n) = t$ and denote by rw_f^* the vector of length 2^t obtained by keeping all nonzero coefficients in rw_f . Then we have the following result.

Corollary 1. *Let $n \in \mathbb{N}^*$ with $w_H(n) = t$. Let $\mathbf{W}$ denote the submatrix formed by the ℓ -th column in $\mathbf{V}^{(t+1)}$ for all integers ℓ satisfying $\ell \preceq n$. Then for any n -variable WAPB function f with $\text{corr}(f) = t$, the 2^t -length vector rw_f^* satisfies the following system of equations in $\mathbf{y} = (y_0, y_1, \dots, y_{2^t-1})^\top \in \{\pm 1\}^{2^t}$:*

$$\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}.$$

The above characterization of t -correctors in terms of a Vandermonde matrix will be used to show the non-existence of WPB functions and WAPB functions of certain corrector order t in the next section.

4 Corrector Order of WAPB Functions

Section 3 characterizes t -correctors in terms of restricted Walsh transforms and Vandermonde matrices. This section will further discuss correction order of WAPB functions by studying the nonzero coefficients in their restricted Walsh transform vectors.

4.1 Non-Existence of 1-resilient WPB Functions

Note that the slice $E_{k,n}$ has even cardinality if and only if $\binom{n}{k}$ is even, equivalently, $k \not\preceq n$ according to Lemma 1. By definition, WPB functions are balanced on each $E_{k,n}$ for any $k \in [1, n-1]$, indicating that they exist only for the case where $n = 2^m$ for certain positive integer m . The following proposition shows the non-existence of 1-resilient WPB functions.

Proposition 1. *Let f be an n -variable WPB function with $n = 2^m$ for $m \in \mathbb{N}^*$. Then $\text{res}(f) = \text{corr}(f) = 0$.*

Proof. By Definition 7, a WPB function is balanced, hence 0-corrector and 0-resilient. Below we show a WPB function cannot be 1-corrector, thus not 1-resilient.

By Theorem 3 we see that $\text{corr}(f) = 1$ if and only if

$$\mathbf{V}^{(2)} \cdot \text{rw}_f = \mathbf{0}_2,$$

where the restricted Walsh vector rw_f satisfies that $\mathcal{W}_{f,\ell} = 0$ for any $\ell \in [1, n-1]$ since f is balanced on $E_{\ell,n}$. Excluding these columns in $\mathbf{V}^{(2)}$ reduces the above system to

$$\begin{bmatrix} 1 & 1 \\ 0 & n \end{bmatrix} \cdot \begin{bmatrix} \mathcal{W}_{f,0} \\ \mathcal{W}_{f,n} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which contradicts the definition of WPB functions in Definition 7. □

4.2 Corrector Order of n -variable WAPB Functions when $w_H(n) = 2$ and $w_H(n) = 3$

We state two propositions for WAPB functions below.

Proposition 2. *Let $n \in \mathbb{N}^*$ with $w_H(n) = 2$. Then any WAPB function f has $\text{corr}(f) \leq 1$. Furthermore, the equality is reached if and only if the restricted Walsh vector rw_f satisfies*

$$\mathcal{W}_{f,\ell} = \mathcal{W}_{f,0} \cdot (-1)^{w_H(\ell)}, \forall \ell \preceq n.$$

Proof. By Corollary 1 it suffices to consider the solution of the linear system $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}$ for $t = 1$ and $t = 2$. More precisely, letting $n = 2^{i_1} + 2^{i_2}$ with positive integers $i_1 < i_2$, we need to study the following two linear systems:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2^{i_1} & 2^{i_2} & (2^{i_1} + 2^{i_2}) \end{bmatrix} \cdot \mathbf{y} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (8)$$

and

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2^{i_1} & 2^{i_2} & (2^{i_1} + 2^{i_2}) \\ 0 & 2^{2i_1} & 2^{2i_2} & (2^{i_1} + 2^{i_2})^2 \end{bmatrix} \cdot \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad (9)$$

where $\mathbf{y} = (y_{00}, y_{01}, y_{10}, y_{11}) \in \{\pm 1\}^4$. It can be verified that the solutions of (8) are given by

$$\mathbf{y} = \pm(1, -1, -1, 1).$$

This is equivalent to saying that $\mathcal{W}_{f,\ell} = \mathcal{W}_{f,0} \cdot (-1)^{\mathbf{w}_H(\ell)}$ for any $\ell \preceq n$. Furthermore, these solutions do not satisfy (9), which implies that there exists no WAPB function f with $\text{corr}(f) = 2$ when $\mathbf{w}_H(n) = 2$. $\square$

Proposition 3. *Let $n \in \mathbb{N}^*$ with $\mathbf{w}_H(n) = 3$. Then any WAPB function f has $\text{corr}(f) \leq 2$. Furthermore, the equality is reached if and only if the restricted Walsh vector rw_f satisfies*

$$\mathcal{W}_{f,\ell} = \mathcal{W}_{f,0} \cdot (-1)^{\mathbf{w}_H(\ell)}, \forall \ell \preceq n.$$

Proof. Similarly as in the proof for Proposition 2, we consider the solutions to $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}^\top$ for $t = 2$ and $t = 3$.

Let $n = 2^{i_1} + 2^{i_2} + 2^{i_3}$ with positive integers $i_1 < i_2 < i_3$, this reduces to studying the solutions for

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2^{i_1} & 2^{i_2} & (2^{i_1} + 2^{i_2}) & 2^{i_3} & (2^{i_1} + 2^{i_3}) & (2^{i_2} + 2^{i_3}) & (2^{i_1} + 2^{i_2} + 2^{i_3}) \\ 0 & 2^{2i_1} & 2^{2i_2} & (2^{i_1} + 2^{i_2})^2 & 2^{2i_3} & (2^{i_1} + 2^{i_3})^2 & (2^{i_2} + 2^{i_3})^2 & (2^{i_1} + 2^{i_2} + 2^{i_3})^2 \end{bmatrix} \cdot \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (10)$$

and

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2^{i_1} & 2^{i_2} & (2^{i_1} + 2^{i_2}) & 2^{i_3} & (2^{i_1} + 2^{i_3}) & (2^{i_2} + 2^{i_3}) & (2^{i_1} + 2^{i_2} + 2^{i_3}) \\ 0 & 2^{2i_1} & 2^{2i_2} & (2^{i_1} + 2^{i_2})^2 & 2^{2i_3} & (2^{i_1} + 2^{i_3})^2 & (2^{i_2} + 2^{i_3})^2 & (2^{i_1} + 2^{i_2} + 2^{i_3})^2 \\ 0 & 2^{3i_1} & 2^{3i_2} & (2^{i_1} + 2^{i_2})^3 & 2^{3i_3} & (2^{i_1} + 2^{i_3})^3 & (2^{i_2} + 2^{i_3})^3 & (2^{i_1} + 2^{i_2} + 2^{i_3})^3 \end{bmatrix} \cdot \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

where $\mathbf{y} = (y_{000}, y_{001}, y_{010}, y_{011}, y_{100}, y_{101}, y_{110}, y_{111}) \in \{\pm 1\}^8$.

Let us first study the solutions of (10). We will iterate through the rows of the matrix to gradually build necessary conditions on its solutions. We refer to Row k as the row indexed by the exponent k for $0 \leq k < 3$ in $\mathbf{W}$.

Row 0 indicates that $\mathbf{y}$ must be a balanced vector, namely, half of its entries equal 1 and the other half equal -1 . Row 1 yields the following equation:

$$\begin{aligned} & (y_{001} + y_{011} + y_{101} + y_{111}) \cdot 2^{i_1} + \\ & (y_{010} + y_{011} + y_{110} + y_{111}) \cdot 2^{i_2} + \\ & (y_{100} + y_{101} + y_{110} + y_{111}) \cdot 2^{i_3} = 0 \end{aligned}$$

For ease of presentation we denote the sum coefficients of $2^{i_1}, 2^{i_2}, 2^{i_3}$ by s_1, s_2, s_3 , respectively. Then the equation from Row 1 becomes

$$s_1 \cdot 2^{i_1} + s_2 \cdot 2^{i_2} + s_3 \cdot 2^{i_3} = 0. \quad (12)$$

By construction, the sums s_1, s_2, s_3 can only take for value any $\pm 4, \pm 2$ or 0. By further applying the constraint of balancedness from Row 0, all possible $\binom{8}{4}$ balanced $\mathbf{y}$ exhibit only the following possible options for the multiset (that allows for same elements) $\{s_1, s_2, s_3\}$:

$$\{0, 0, 0\}, \{0, 0, \pm 2\}, \{0, \pm 2, \pm 2\}, \{\pm 2, \pm 2, \pm 2\}, \{0, 0, \pm 4\}.$$

This means that each term on the left hand of Eq. (12) is either zero or a power of 2. Since $i_1 < i_2 < i_3$, it implies that Eq. (12) can be true only when $s_1 = s_2 = s_3 = 0$.

From Row 2 one gets the following equation (after re-arranging the terms):

$$\begin{aligned} & (y_{001} + y_{011} + y_{101} + y_{111}) \cdot 2^{2i_1} + \\ & (y_{010} + y_{011} + y_{110} + y_{111}) \cdot 2^{2i_2} + \\ & (y_{100} + y_{101} + y_{110} + y_{111}) \cdot 2^{2i_3} + \\ & (y_{011} + y_{111}) \cdot 2 \cdot 2^{i_1+i_2} + \\ & (y_{101} + y_{111}) \cdot 2 \cdot 2^{i_1+i_3} + \\ & (y_{110} + y_{111}) \cdot 2 \cdot 2^{i_2+i_3} = 0. \end{aligned} \quad (13)$$

After applying the preceding constraints, namely, balancedness of $\mathbf{y}$ from Row 0 and $s_1 = s_2 = s_3 = 0$ from Row 1, we obtain

$$2s_{1,2} \cdot 2^{i_1+i_2} + 2s_{1,3} \cdot 2^{i_1+i_3} + 2s_{2,3} \cdot 2^{i_2+i_3} = 0,$$

where $s_{1,2}, s_{1,3}, s_{2,3}$ correspond to the sum coefficient of $2^{i_1+i_2}, 2^{i_1+i_3}$ and $2^{i_2+i_3}$, respectively. We now discuss the values of $s_{1,2}, s_{1,3}$ and $s_{2,3}$. Assume $s_{1,2} = \pm 2$. Then $y_{011} = y_{111} = \pm 1$.

By considering both the condition of balancedness and that $s_1 = s_2 = s_3 = 0$, we necessarily have $y_{101} = y_{110} = -y_{011} = -y_{111}$. This does not satisfy Eq. (13). The same logic applies if $y_{101} = y_{111}$ or if $y_{110} = y_{111}$. The only combination that works is $y_{011} = y_{101} = y_{110} = -y_{111}$, thereby giving the following solutions to Eq. (13)

$$\mathbf{y} = \pm(1, -1, -1, -1, 1, 1, 1, -1).$$

Furthermore, these solutions do not satisfy the last equation in (11). That is to say, there is not WAPB function f with $\text{corr}(f) = 3$ when $w_H(n) = 3$. $\square$

4.3 Corrector Order of n -variable WAPB Functions for $w_H(n) > 3$

We shall consider general cases by following the discussion for the cases where $w_H(n) = 2$ and $w_H(n) = 3$. In order to discuss the linear systems in a unified way, we need the following lemma.

Lemma 2. Assume $n \in \mathbb{N}^*$ has $\text{supp}(n) = \{i_1, i_2, \dots, i_t\}$ and $x_s = 2^{i_s}$ for $s = 1, 2, \dots, t$. For $1 \leq u \leq t$, the equation $\sum_{\ell \preceq n} \ell^u y_\ell = 0$ can be re-written as

$$\sum_{s=1}^t \sum_{\substack{\{j_1, \dots, j_s\} \subseteq [t] \\ u_{j_1} + \dots + u_{j_s} = u \\ u_{j_1}, \dots, u_{j_s} \geq 1}} \frac{u!}{u_{j_1}! \dots u_{j_s}!} \left(\sum_{\substack{\{i_{j_1}, \dots, i_{j_s}\} \subseteq \text{supp}(\ell) \\ \ell \preceq n}} y_\ell \right) x_{j_1}^{u_{j_1}} \dots x_{j_s}^{u_{j_s}} = 0. \quad (14)$$

Proof. Note that the expansion of $(x_1 + \dots + x_s)^u$ is given by

$$(x_1 + \dots + x_s)^u = \sum_{\substack{u_1 + \dots + u_s = u \\ u_1, \dots, u_s \geq 0}} \frac{u!}{u_1! \dots u_s!} x_1^{u_1} \dots x_s^{u_s},$$

where $\frac{u!}{u_1! \dots u_s!}$ is the multinomial coefficient. In the above sum $(u_1, \dots, u_s)$ can be seen as a non-negative partition of the integer u . For simplicity, we denote by $P_s(u)$ the set of all such partitions of u . Observe that for any $(u_1, \dots, u_s) \in P_s(u)$, the integers $u_1, \dots, u_s, 0$ form an $(s+1)$ -tuple in $P_{s+1}(u)$. This implies the inclusive relation

$$P_s(u) \subset P_{s+1}(u), \quad s = 1, 2, \dots$$

On the other hand, for an $(s+1)$ -tuple $(y_1, \dots, y_{s+1})$ in $P_{s+1}(u)$ with (at least) one zero coordinate, it will degenerate to an s -tuple in $P_s(u)$ after one zero coordinate is removed. Hence, for $s \leq u$, we can partition the set $P_s(u)$ as

$$P_s(u) = P_s^*(u) \sqcup P_{s-1}^*(u) \sqcup \dots \sqcup P_1^*(u),$$

where $P_r^*(u)$ contains those r -tuples $(u_1, \dots, u_r)$ with $u_1, \dots, u_r \geq 1$.

Now we consider the expansion of the sum $\sum_{\ell \preceq n} y_\ell \ell^u = 0$ with $n = 2^{i_1} + \dots + 2^{i_t}$ and $u > 0$. By denoting $x_j = 2^{i_j}$ and re-arranging the terms in its expansions, we obtain

$$\begin{aligned} \sum_{\ell \preceq n} \ell^u y_\ell &= \sum_{s=1}^t \sum_{\substack{\text{supp}(\ell) = \{i_{j_1}, \dots, i_{j_s}\} \\ \{j_1, \dots, j_s\} \subseteq [t]}} (x_{j_1} + \dots + x_{j_s})^u y_\ell \\ &= \sum_{s=1}^t \sum_{\substack{\text{supp}(\ell) = \{i_{j_1}, \dots, i_{j_s}\} \\ \{j_1, \dots, j_s\} \subseteq [t]}} y_\ell \sum_{(u_1, \dots, u_s) \in P_s(u)} \frac{u!}{u_{j_1}! \dots u_{j_s}!} x_{j_1}^{u_{j_1}} \dots x_{j_s}^{u_{j_s}} \\ &= \sum_{s=1}^t \sum_{\substack{\text{supp}(\ell) = \{i_{j_1}, \dots, i_{j_s}\} \\ \{j_1, \dots, j_s\} \subseteq [t] \\ (u_{j_1}, \dots, u_{j_s}) \in P_s(u)}} \frac{u!}{u_{j_1}! \dots u_{j_s}!} y_\ell x_{j_1}^{u_{j_1}} \dots x_{j_s}^{u_{j_s}} \end{aligned} \quad (15)$$

where the last equality is obtained by re-arranging the summands for $s = 1, \dots, t$ according to the partition $P_s(u) = P_s^*(u) \sqcup P_{s-1}^*(u) \sqcup \dots \sqcup P_1^*(u)$. $\square$

Example 1. Let $\text{supp}(n) = \{i_1, i_2, i_3\}$, $u = 2$ and $x_t = 2^{i_t}$ for $t = 1, 2, 3$. Then,

$$\begin{aligned}
\sum_{\ell \preceq n} \ell^2 y_\ell &= \sum_{s=1}^t \sum_{\substack{\{j_1, \dots, j_s\} \subseteq [t] \\ (u_{j_1}, \dots, u_{j_s}) \in P_s^*(u)}} \binom{u}{u_{j_1}, \dots, u_{j_s}} \left(\sum_{\substack{\{i_{j_1}, \dots, i_{j_s}\} \subseteq \text{supp}(\ell) \\ \ell \preceq n}} y_k \right) x_{j_1}^{u_1} \dots x_{j_s}^{u_{j_s}} \\
&= \sum_{\substack{j \in [3] \\ u_j = 2}} \binom{2}{u_j} \left(\sum_{\substack{\{i_j\} \subseteq \text{supp}(\ell) \\ \ell \preceq n}} y_\ell \right) x_j^2 \\
&\quad + \sum_{\substack{j_1, j_2 \in [3] \\ u_{j_1} = u_{j_2} = 1}} \binom{2}{u_{j_1}, u_{j_2}} \left(\sum_{\substack{\{i_{j_1}, i_{j_2}\} \subseteq \text{supp}(\ell) \\ \ell \preceq n}} y_\ell \right) x_{j_1} x_{j_2} \\
&= \sum_{j=1}^3 \left(\sum_{2^{i_j} \preceq \ell \preceq n} y_\ell \right) x_j^2 + 2 \sum_{j_1, j_2 \in [3]} \left(\sum_{2^{i_{j_1} + i_{j_2}} \preceq \ell \preceq n} y_\ell \right) x_{j_1} x_{j_2},
\end{aligned}$$

where the second equality has only two summations. Denoting y_ℓ for $\ell = b_1 2^{i_1} + b_2 2^{i_2} + b_3 2^{i_3}$ as $y_{b_1 b_2 b_3}$ with $b_1, b_2, b_3 \in \{0, 1\}$, we obtain the following expansion

$$\begin{aligned}
\sum_{k \preceq n} y_k k^2 &= (y_{100} + y_{110} + y_{101} + y_{111}) x_1^2 + (y_{010} + y_{110} + y_{011} + y_{111}) x_2^2 \\
&\quad + (y_{001} + y_{101} + y_{011} + y_{111}) x_3^2 \\
&\quad + 2((y_{110} + y_{111}) x_1 x_2 + (y_{101} + y_{111}) x_1 x_3 + (y_{011} + y_{111}) x_2 x_3)
\end{aligned}$$

where $x_1 = 2^{i_1}, x_2 = 2^{i_2}, x_3 = 2^{i_3}$.

Below we shall use the representation of Lemma 2 to investigate the corrector order of n -variable WAPB functions for infinitely many integers n .

Recall that a superincreasing sequence of integers $\mathbf{s} = [s_1, s_2, s_3, \dots]$ satisfies $s_{r+1} > \sum_{i=1}^r s_i$ for all integers $r \geq 1$. Every subset sum of a superincreasing sequence is unique. Furthermore, we shall call $\mathbf{s}$ an m -gap superincreasing sequence if $\mathbf{s}$ is a superincreasing sequence and

$$\min \left\{ \left| \sum_{i=1}^h a_i - \sum_{i=1}^h a'_i \right| : (a_1, \dots, a_h) \neq (a'_1, \dots, a'_h) \in A^h, h \geq 1 \right\} = m, \quad (16)$$

where A is the set of all entries in $\mathbf{s}$.

Theorem 4. Let $n \in \mathbb{N}^*$ with $\text{supp}(n) = \{i_1, \dots, i_t\}$ with $i_1 < \dots < i_t$ and $t > 3$. Suppose $[i_1, \dots, i_t]$ forms a t -gap superincreasing sequence. Then any n -variable WAPB function f has $\text{corr}(f) \leq t - 1$. Furthermore, the equality is reached if and only if the restricted Walsh vector rw_f satisfies

$$\mathcal{W}_{f, \ell} = \mathcal{W}_{f, 0} \cdot (-1)^{\mathbf{w}_H(\ell)}, \forall \ell \preceq n.$$

Proof. From Theorem 3 we study the correction order of such functions by considering solutions to the linear systems $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_t$ and $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}$ where $\mathbf{y} = (y_{0\dots 00}, y_{0\dots 01}, y_{0\dots 10}, \dots, y_{1\dots 10}, y_{1\dots 11}) \in \{\pm 1\}^{2^t}$.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 0 & 2^{i_1} & 2^{i_2} & (2^{i_1} + 2^{i_2}) & \dots & (2^{i_1} + \dots + 2^{i_t}) \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 2^{(t-1)i_1} & 2^{(t-1)i_2} & (2^{i_1} + 2^{i_2})^{t-1} & \dots & (2^{i_1} + \dots + 2^{i_t})^{t-1} \end{bmatrix} \cdot \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (17)$$

For each y_i we can write $i = \sum_{\delta=0}^{t-1} i_\delta 2^\delta$. Let us introduce the notation $\mathbf{y}_\Delta = \{y_i \mid i_\delta = 1, \forall \delta \in \Delta\}$ for some $\Delta \subseteq \{0, 1, \dots, t-1\}$. Setting the starting row of $\mathbf{W}$ from zero, we will proceed by induction on Rows 1 to Row $t-1$ of $\mathbf{W}$. We want to prove that all sum coefficients must be null for each linear relation induced by the first r rows. Row 0 forces $\mathbf{y}$ to be a balanced vector over $\{\pm 1\}$.

Initialization.

Let $r = 1$. Row 1 induces the equation below

$$\left(\sum_{y_i \in \mathbf{y}_{\{1\}}} y_i \right) \cdot 2^{i_1} + \left(\sum_{y_i \in \mathbf{y}_{\{2\}}} y_i \right) \cdot 2^{i_2} + \dots + \left(\sum_{y_i \in \mathbf{y}_{\{t\}}} y_i \right) \cdot 2^{i_t} = 0.$$

For ease of presentation we denote the sum coefficients of 2^{i_j} by s_j . Then, the equation from Row 1 becomes

$$s_1 \cdot 2^{i_1} + s_2 \cdot 2^{i_2} + \dots + s_t \cdot 2^{i_t} = 0. \quad (18)$$

Since $\mathbf{y}$ is balanced, it is clear that any s_j can take for values any even number or its opposite in the range $[0, 2^{t-1}]$ because $|\mathbf{y}_\Delta| = 2^{t-|\Delta|}$. From the initial condition, $[i_1, \dots, i_t]$ is a t -gap superincreasing sequence and since $s_i < 2^t$ and because the gap between any two i_j is at least t , there does not exist any $s_j \neq 0$ satisfying Eq. (18).

Let $r = 2$. From Row 2 one gets the following equation after re-arranging the terms

$$\begin{aligned} & \left(\sum_{y_i \in \mathbf{y}_{\{1\}}} y_i \right) \cdot 2^{2i_1} + \left(\sum_{y_i \in \mathbf{y}_{\{2\}}} y_i \right) \cdot 2^{2i_2} + \dots + \left(\sum_{y_i \in \mathbf{y}_{\{t\}}} y_i \right) \cdot 2^{2i_t} \\ & + 2 \left(\sum_{y_i \in \mathbf{y}_{\{1,2\}}} y_i \right) \cdot 2^{i_1+i_2} + \dots + 2 \left(\sum_{y_i \in \mathbf{y}_{\{t-1,t\}}} y_i \right) \cdot 2^{i_{t-1}+i_t} = 0. \end{aligned}$$

We use the previously defined notations and knowledge that $s_1 = s_2 = \dots = s_t = 0$ to simplify it into

$$2s_{1,2} \cdot 2^{i_1+i_2} + 2s_{1,3} \cdot 2^{i_1+i_3} + \dots + 2s_{t-1,t} \cdot 2^{i_{t-1}+i_t} = 0. \quad (19)$$

Similarly as above, we see that any s_j can take for values any even number or its opposite in the range $[0, 2^{t-2}]$ and we make use of the initial condition that $[i_1, \dots, i_t]$ is a t -gap superincreasing sequence and bring the same argument to prove there does not exist any $s_{i,j} \neq 0$ satisfying Eq. (19).

Induction: we shall prove that if the statement is true for $r-1$, then it is true for r , $r \leq t-1$. From Row r and using Lemma 2 one has

$$\sum_{\ell \leq n} \ell^r y_\ell = \sum_{m=1}^t \sum_{\substack{\{j_1, \dots, j_m\} \subseteq [t] \\ r_{j_1} + \dots + r_{j_m} = r \\ r_{j_1}, \dots, r_{j_m} \geq 1}} \frac{r!}{r_{j_1}! \dots r_{j_m}!} \left(\sum_{\substack{\{i_{j_1}, \dots, i_{j_m}\} \subseteq \text{supp}(\ell) \\ \ell \leq n}} y_\ell \right) x_{j_1}^{r_{j_1}} \dots x_{j_m}^{r_{j_m}} = 0.$$

Let us denote the multinomial coefficient by $\mu_{1, \dots, m} = \frac{r!}{r_{j_1}! \dots r_{j_m}!}$ and the previously defined notations $s_{1, \dots, m} = \sum(y_\ell)$. We rewrite the above equation into its rearranged form

$$\mu_1(s_1 2^{i_1} + \dots + s_t 2^t) + \mu_{1,2}(s_{1,2} 2^{i_1+i_2} + \dots + s_{t-1,t} 2^{i_{t-1}+i_t}) + \dots + \mu_{1,2,\dots,r}(s_{1,2,\dots,r} 2^{i_1+i_2+\dots+i_r} + \dots + s_{t-r,t-r+1,\dots,t} 2^{i_{t-r}+i_{t-r+1}+\dots+i_t}) = 0.$$

We know from Row $r - 1$ that $s_1, \dots, s_t, s_{1,2}, \dots, s_{t-1,t}, \dots, s_{t-(r-1),\dots,t-1,t} = 0$. We then reformulate the above equation into

$$s_{1,2,\dots,r} 2^{i_1+i_2+\dots+i_r} + \dots + s_{t-r,t-r+1,\dots,t} 2^{i_{t-r}+i_{t-r+1}+\dots+i_t} = 0. \quad (20)$$

In a similar fashion, we see that any s_j can take for values any even number or its opposite in the range $[0, 2^{t-r}]$ and we make use of the initial condition that $[i_1, \dots, i_t]$ is a t -gap superincreasing sequence and bring the same argument to prove there does not exist any $s_j \neq 0$ satisfying Eq. (20).

Finally, expanding $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}$ yields $s_{1,\dots,t} = y_{1,\dots,t}$ that cannot satisfy Eq. (17) with knowledge of previous s_j , bringing the argument of impossibility for such a t -corrector. As such, the restricted Walsh vector is uniquely determined by $\mathcal{W}_{f,0}(-1)^{\mathbf{w}_H(\ell)}, \forall \ell \preceq n$. This concludes the proof. $\square$

Based on the different theoretic results of this section such as Proposition 4 and experimental results (Section 5), we introduce the following conjecture on the corrector and resilience order of WAPB functions.

Conjecture 1. Let $n \in \mathbb{N}^*$ with $\mathbf{w}_H(n) = t$. Then there exist WAPB functions f with $\text{corr}(f) = t - 1$ and the restricted Walsh transforms rw_f satisfy

$$\mathcal{W}_{f,\ell} = (-1)^{\mathbf{w}_H(\ell)}, \forall \ell \preceq n.$$

Furthermore, there exists no WAPB function f with $\text{corr}(f) = t$.

Remark 1. Regarding the resilience upper bound of $\mathbf{w}_H(n) - 1$, we know that for all n there exist WAPB functions with resilience order $\mathbf{w}_H(n) - 1$: when we write n as $n = 2^{i_1} + 2^{i_2} + \dots + 2^{i_t}$ where $i_1 < \dots < i_t$, the direct sum of WPB functions in 2^{i_s} variables [49] have resilience order exactly $\mathbf{w}_H(n) - 1$. Such WAPB functions are exactly characterized in the conjecture, namely, their Walsh transforms rw_f are given by $\mathcal{W}_{f,\ell} = (-1)^{\mathbf{w}_H(\ell)}, \forall \ell \preceq n$.

Remark 2. Recall that the Prouhet-Tarry-Escott problem [44] for a positive integer r asks for two disjoint multi-sets S_0 and S_1 of the same number of integers satisfying

$$\sum_{a \in S_0} a^s = \sum_{b \in S_1} b^s, \quad s = 0, 1, \dots, r.$$

The Prouhet-Tarry-Escott problem can be phrased as a r -regular partition of an integer set [3]. More concretely, for a set S of even number of integers, a partition $\{S_0, S_1\}$ of S , i.e., $S_0 \cap S_1 = \emptyset$ and $S_0 \cup S_1 = S$, is r -regular if

$$\sum_{a \in S_0} a^s = \sum_{b \in S_1} b^s, \quad s = 0, 1, \dots, r.$$

When the set $S = \{0, 1, \dots, 2^t - 1\}$, the Prouhet-Thue-Morse sequence [1] gives a $(t - 1)$ -regular partition of S as

$$S_i = \{0 \leq a < 2^t - 1 : \mathbf{w}_H(a) \equiv i \pmod{2}, i = 1, 2\}.$$

Furthermore, in the second part of [3, Conjecture 5.3], the above partition $\{S_0, S_1\}$ was conjectured to be the unique $(t-1)$ -regular partition of $S = \{0, 1, \dots, 2^t - 1\}$. One can easily verify that this partition is not t -regular.

Observe that Conjecture 1 is a generalized version of the second part of Conjecture 5.3 in [3]. For a positive integer n of the form $n = 2^{i_1} + 2^{i_2} + \dots + 2^{i_t}$ with $i_1 < \dots < i_t$, if in Corollary 1 the system of equations $\mathbf{W} \cdot \mathbf{y} = \mathbf{0}_{t+1}$ has a solution $\mathbf{y} = (y_0, y_1, \dots, y_{2^t-1})^\top \in \{\pm 1\}^{2^t}$, then it can be used to derive a partition $\{S_0, S_1\}$ of the set

$$S = \{a_1 2^{i_1} + a_2 2^{i_2} + \dots + a_t 2^{i_t} : a_1, a_2, \dots, a_t \in \mathbb{F}_2\},$$

where S_0, S_1 is derived based on the sign of y_a , where $a = a_1 + 2a_2 + \dots + 2^{t-1}a_t$. That is to say, the existence of a WAPB function f with $\text{corr}(f) = t$ implies the existence of a t -regular partition of the set S .

5 Experimental Results

As illustrations, we provide an open-source implementation validating our main results in SageMath [43] available at <https://www.github.com/wapbres/wapbres>. We give a basis of code to test Theorems 1, 2 and 3 for small parameters. We also verified that Conjecture 1 holds for all dimensions n up to $n = 62$. Our approach consists in exhausting all balanced vectors $\mathbf{y} \in \{\pm 1\}^{\text{w}_H(n)}$ and computing t as in Theorem 3. Precisely there are $\binom{2^{\text{w}_H(n)} - 1}{2^{\text{w}_H(n)} - 1}$ such vectors for which we keep track of the the largest t -corrector. We give a brief summary of the performance of our method in Table 1. All execution times are in seconds and performed on a single core 13th Gen Intel(R) Core(TM) i5-1345U 4.7 GHz. We see that for values of n with $\text{w}_H(n) = 5$, searching through roughly 2^{32} vectors takes around 2h30, indicating the infeasibility of verifying the case of Hamming weight 6 which induces a search space of roughly 2^{64} vectors.

Table 1. Correction order of WAPB functions in $n \leq 62$ variables.

$\text{w}_H(n)$	1	2	3	4	5
$\text{corr}(f)$	0	1	2	3	4
time (s)	0.02	0.03	0.06	0.21	8916.26

6 Conclusion

This article investigated the resilience order of WAPB functions from the perspective of their corrector order. Our main contributions are twofold. First, we give a characterization of the corrector orders of a Boolean function by expressing its restricted Walsh transform vector in terms of linear equations over integers. We established an interesting algebraic relation between the Krawchouk matrix and the Vandermonde matrix, which transforms the linear equations into new ones associated with certain Vandermonde matrix. Second, our analysis of those new equations showed that for infinitely many integers n , the maximum corrector order of n -variable WAPB functions is tightly upper bounded by the Hamming weight of n .

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